Topic B

Multiple choice questions

1 The side of a solid cube of mass M is d. The density of the cube is ρ. The side of another cube of the same material is $2d$. What is the mass and density of this cube?

	Mass	Density
A	$2M$	$\frac{\rho}{2}$
B	$2M$	ρ
C	$8M$	$\frac{\rho}{2}$
D	$8M$	ρ

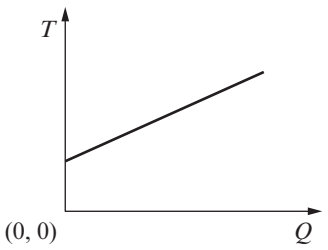
2 The temperature of a substance is changed from −32 °C to −46 °C. What is the change in the temperature of the substance in degrees kelvin?

A 14
B −14
C 259
D 287

3 Molecules of mass of order 10^{-26} kg are kept at temperature 300 K. What is an order of magnitude estimate of the average speed of the molecules?

A $10\ \text{m s}^{-1}$
B $10^2\ \text{m s}^{-1}$
C $10^3\ \text{m s}^{-1}$
D $10^4\ \text{m s}^{-1}$

4 The graph shows the variation of the temperature T of a solid as it is being heated with energy Q.



In order to determine the specific heat capacity of the solid, it is suggested that the following quantities be measured:

I the mass of the solid
II the gradient of the graph
III the vertical intercept of the graph.

What is needed in order to determine the specific heat capacity?

A I and II
B I and III
C II and III
D I, II and III.

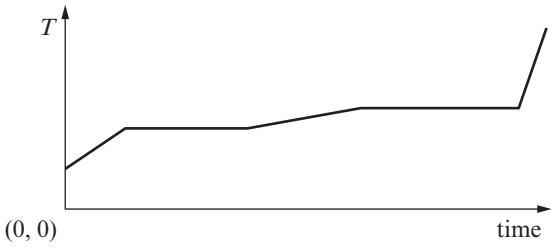
5 Body X, with temperature 0 °C, is brought into thermal contact with body Y of equal mass and of temperature 100 °C. The only exchanges of heat that take place are between X and Y. The specific heat capacity of X is greater than that of Y. Which statement about the final equilibrium temperature T of the two bodies is correct?

A $T = 50$ °C
B $0 < T < 50$ °C
C 100 °C $> T >$ 50 °C
D Answer depends on value of mass.

6 Energy is provided to a liquid at its boiling point at a rate of P joules per second. The rate at which mass is boiling away is μ kg per second. What is the specific latent heat of vaporisation of the liquid?

A μP
B $\frac{P}{\mu}$
C $\frac{\mu}{P}$
D $\frac{1}{\mu P}$

7 The graph shows the variation with time of the temperature T of a material as it is brought from the solid state into the liquid state and then into the vapour state. Energy is provided at a constant rate.



Which is the correct comparison of the specific heat capacities c_S, c_L and c_V in the solid, liquid and vapour states?

A $c_V > c_S > c_L$
B $c_S > c_L > c_V$
C $c_V > c_L > c_S$
D $c_L > c_S > c_V$

8 5 kg of water at 10 °C are mixed with 2 kg of water at 80 °C. What is the final temperature of the water?

A 30 °C
B 45 °C
C 55 °C
D 70 °C

9 Equal quantities of liquids X and Y are mixed. X is at 20 °C, and Y is at 80 °C. The specific heat capacity of X is double that of Y. What is the final temperature of the mixture?

A 40 °C
B 50 °C
C 60 °C
D 70 °C